

BANTROBUSA KAZIBWE

Backend Developer (.NET & Fintech) | Kampala, Uganda | EAT (UTC+3)
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PROFESSIONAL PROFILE

I'm a backend developer based in Kampala, Uganda, with several years of hands-on experience building production backend systems and REST APIs, including in C# and .NET Core. My work covers the full development lifecycle, from early architecture and API design through to live deployment, monitoring, and ongoing maintenance, with a consistent focus on transaction accuracy, data integrity, and reliability.

I've built systems for some of Uganda's most important institutions, including the National Social Security Fund and the Uganda Civil Aviation Authority, and I've worked in collaborative, multi-disciplinary teams both on-site and remote. My backend work spans C# and .NET Core, Node.js and Express, Laravel, and FastAPI, alongside hands-on fintech experience building SandiAI, a savings and mobile money platform handling real transactions, payment integrations, and credit scoring for East African users. I bring the same discipline around clean REST API design, database architecture, and third-party integration to every stack I work in.

I've built and maintained REST APIs, integrated third-party and legacy systems, and shipped production systems that real users depend on every day, including live fintech products handling actual money movement. I own products end to end, from architecture through deployment and production support, including SandiAI and DHQ AI at Ketii AI and Hoscover at Jurya Technologies. I'm looking for a backend opportunity in fintech where I can bring that same ownership to transaction processing, integration, and system reliability.

TECHNICAL SKILLS

Backend & API Development: C# and .NET Core for enterprise backend services and legacy system data aggregation; Node.js and Express with TypeScript for RESTful API design and backend services; Laravel (PHP) for RESTful API design, modular MVC architecture, multi-role authentication, and service-layer patterns; FastAPI (Python) for lightweight high-performance APIs; Java for additional backend contexts

API Protocols & Security: REST APIs and JSON; SOAP APIs, including studying the protocol and building an integration adapter to connect Hoscover with a hospitality property management system (PMS) that communicated over SOAP; JWT-based authentication implemented across PACT, SandiAI, and Hoscover

Real-Time and Streaming: WebSockets, REST APIs, streaming data pipelines, live data integration across distributed product ecosystems

Third-Party & System Integrations: Africa's Talking (SMS), PesaPal (payments), and Voiceflow (conversational AI) integrated into production systems; WhatsApp Business API pipeline awareness; IoT sensor data feeds; government legacy system API aggregation

Databases: PostgreSQL (via Supabase), MySQL, Firebase Firestore, MariaDB, SQLite. Comfortable with schema design, query optimisation, and selecting the appropriate data layer for the project's scale and requirements

Frontend Frameworks: React, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Bootstrap

Testing and Documentation: PHPUnit for backend test coverage; Swagger/OpenAPI for API contract documentation; user guides, developer onboarding materials, and codebase documentation

Developer Tooling: Git and GitHub for version control and collaborative branching workflows; Vercel and Railway for deployment pipelines

Backend Deployment: Railway for backend service deployment and hosting; Docker for containerizing services when Railway's native build doesn't support the stack, including lightweight C# services; Vercel for frontend continuous deployment; experience managing live production environments

Mobile Development: Flutter and Dart for cross-platform iOS and Android applications; React Native for cross-platform mobile development; Firebase Authentication, Firestore real-time databases, Cloud Messaging, and Analytics; Clean Architecture; state management with BLoC, Provider, and GetX

OTA Mobile Updates: Shorebird for Flutter over-the-air (OTA) app updates, managing and deploying live patches to mobile applications without requiring full store releases; Expo for React Native development, builds, and OTA updates

Data Visualisation: Recharts, Chart.js. Built production dashboards aggregating real-time data from multiple sources

AI and Machine Learning: Automatic Speech Recognition using Whisper and Sunbird AI models; on-device inference with TensorFlow Lite integrated into Flutter applications; backends in Python and C#/.NET that automatically generate structured analytical outputs from raw data; LLM API integration including Anthropic Claude and OpenAI

Design and Handoff: Figma for UI design, prototyping, wireframing, and developer handoff. Able to implement designs faithfully and contribute design opinions constructively

HOW I USE AI IN MY DAILY WORKFLOW

AI is a real part of how I work day to day, but I use it with clear boundaries around what it's good at and where it shouldn't be trusted.

- Error diagnosis and debugging support: I use AI assistants as a first-pass diagnostic tool. I describe the problem and share the relevant code, then use the response to generate hypotheses that I verify myself. It speeds up debugging without cutting corners on actually understanding what went wrong.
- UI enhancement and interface feedback: I use AI for rapid feedback on interface decisions, colour contrast, layout hierarchy, and component structure, functioning like a design review partner when I am working alone.
- Peer-style code review: I paste sections of my own code into AI assistants and ask for critique the way I would ask a senior colleague, looking for readability issues, edge cases, and naming inconsistencies, and treating the output as a starting point rather than a verdict.
- What I do not use AI for: I do not pass API keys, authentication credentials, core backend logic, or any sensitive user data through AI tools. I handle those parts of the work directly, with full awareness of what is at stake when those boundaries are not respected.

PROJECT EXPERIENCE AND DELIVERY

SandiAI | *Keti AI (Client Project)*

SandiAI is a savings circle (VSLA) platform targeting East African markets, particularly Uganda, that I have built as a full-stack developer across every layer of the product: web application, mobile application, backend, and live deployment.

- Built and deployed the SandiAI web application, giving users and administrators a browser-based interface for managing savings groups and transactions.
- Developed the SandiAI mobile application in React Native (Expo), currently live on the Google Play Store for testing ahead of production release.
- Architected the backend using Express and TypeScript with Prisma and Supabase for data modelling and storage, implementing JWT-based authentication and integrating Africa's Talking for SMS, PesaPal for payments, and Voiceflow for conversational features.
- Own the full deployment pipeline, managing live production infrastructure for both the web and backend services.

Live: <https://sandi-ai-web-app-production.up.railway.app> | Google Play: <https://play.google.com/store/apps/details?id=com.sandiai.app>

DHQ AI | *Keti AI (Client Project)*

DHQ AI is a nationwide school attendance system I built for Keti AI, tracking teacher and student attendance across rural schools in Uganda through a structured onboarding flow: district field officers onboard head teachers, and head teachers onboard their teachers.

- Built a geofenced attendance system for teachers, where a head teacher generates a session code that only updates the attendance record when entered from within the school premises.
- Built student attendance tracking through two methods: uploading a photo of the attendance sheet or marking attendance directly from a student's profile.
- Built the frontend and mobile app for the platform, with backend support to bring the full system together.

Hoscover | *Full-Stack Developer & Team Lead* | *Jurya Technologies* | 2025 – 2026

Hoscover is a hospitality technology platform I built as Full-Stack Developer and Team Lead for Jurya Technologies, architecting and developing it end-to-end. This project was one of the most formative experiences of my career, requiring me to think holistically across system architecture, user experience, API design, frontend development, and production deployment simultaneously. I was responsible for the complete technical lifecycle of the product, from its earliest design decisions through to its live deployment on Vercel.

- Designed and implemented a data-rich frontend dashboard that aggregates live data streams and presents them as clear, decision-ready interfaces.
- Architected the full backend API layer and integrated it with the frontend, managing real-time data flows across the system.
- Studied SOAP and built an integration adapter to connect Hoscover with a hospitality property management system (PMS) that communicated over SOAP.
- Made all major technology and architecture decisions independently, balancing speed of delivery against long-term maintainability.
- Led the development team as Team Lead, producing a developer onboarding guide to help new engineers get up to speed quickly.
- Deployed and maintained the live production environment on Vercel, managing continuous deployment workflows and monitoring live behaviour.
- Built the authentication and session-handling flow using JWT, including automatic redirect-to-login on session expiry that returns users to their intended page after they sign back in.

Live: <https://hoscover-frontend-snowy.vercel.app> | Session Handling Example: <https://hoscover-frontend-snowy.vercel.app/login?reason=session-expired&redirect=%2Fdashboard>

Cross-Platform Mobile Application | PACT Consultancy | September 2025 – May 2026

At PACT Consultancy I worked as a Software Engineer on a hybrid basis, contributing to the development and ongoing maintenance of cross-platform mobile applications built in Flutter.

- Developed and maintained Flutter mobile applications targeting both Android and iOS, with a strong emphasis on consistent behaviour and reliability across platforms.
- Implemented over-the-air (OTA) update delivery using Shorebird, enabling live patches to be pushed to users' devices without a full store release cycle. This cut the time between a fix being ready and users receiving it, and gave the team much greater agility in responding to production issues.
- Handled backend integration, including JWT-based authentication, platform-specific debugging, and performance optimisation across the application stack.
- Worked closely with a multi-disciplinary team where reliability and user satisfaction were held as central priorities.

Web App: <https://pactorg.com> | Deployed App: <https://app.pactorg.com> | Google Play: <https://play.google.com/store/apps/details?id=com.pact.workflow>

Web-Based Survey Application | Uganda Civil Aviation Authority via NAJOD Technologies | March – June 2025

Contracted through NAJOD Technologies, I developed a fully production-ready web-based survey application for the Uganda Civil Aviation Authority, one of Uganda's most important regulatory bodies. This was a high-stakes engagement where reliability, accessibility, and data integrity were non-negotiable.

- Built the system in Laravel with a strong emphasis on long-term maintainability, using modular controllers and clearly separated service layers.
- Implemented trilingual support across English, Luganda, and Kiswahili using Laravel Localization, making the platform accessible to a significantly broader range of respondents.
- Delivered the project in two-week agile sprints in direct collaboration with survey teams and data analysts, adapting to evolving requirements without compromising delivery timelines.
- Maintained PHPUnit test coverage throughout the project to protect reliability and catch regressions before they reached production.

Live: <https://ucaa-survey.hippozug.com>

Internal Systems Development | National Social Security Fund Uganda | June – August 2025

During my internship at NSSF Uganda I worked concurrently on two significant internal systems, both with direct operational impact on how the organisation functioned day to day.

- Built a service helpdesk application in Laravel to replace a third-party vendor platform, incorporating multi-department and multi-role user management fully integrated with NSSF's existing authentication infrastructure.
- Developed a .NET Core backend in C# that connected to two separate legacy system APIs, handling authentication against each and normalising their different data formats into one consistent structure. Aggregated this data into a centralised analytics layer exposed through REST endpoints, powering a unified real-time dashboard that replaced manual, per-department reporting with a single live view.

Both projects required navigating complex existing infrastructure with care, communicating clearly with non-technical stakeholders, and delivering working software within a defined internship window.

FarmEase Agricultural Management System | *Ndejje University FarmEase Initiative* | February – March 2025

FarmEase was a Flutter and Firebase agricultural management system built for the Ndejje University FarmEase Initiative, giving me hands-on experience solving real operational problems through mobile software.

- Led full development from initial architecture through delivery, incorporating QR code scanning and generation for unique livestock tagging, real-time updates to animal health and production records through Firestore, and a C#/.NET backend that automatically generated farm operational reports.
- Collaborated with IoT engineers on prototype sensor integrations to capture environmental data from the farm, connecting physical infrastructure to the digital management layer.
- Integrated third-party APIs for payment processing and SMS notifications, extending the system's utility beyond simple record-keeping.
- Delivered hands-on training sessions to farm staff alongside thorough user documentation, ensuring the system was actually used rather than simply handed over.

Crest AI | *Speech Recognition and Translation* | May – July 2025

Working remotely with the Crest AI team, I contributed as a full-stack developer across both frontend and backend workstreams on a multilingual speech translation product.

- Built FastAPI endpoints to power multilingual and live speech translation features, designing clean, well-documented API contracts for frontend integration.
- Constructed an Automatic Speech Recognition engine using the Whisper model alongside Sunbird AI, gaining direct, hands-on experience with what production ML systems require in terms of robustness, latency, and error handling.

Lattelink Social Platform | | 2024 – 2025

I designed and built a social media platform in Flutter, taking the product from initial concept through to a working, deployable application.

- Implemented real-time messaging through Firestore streams, multimedia sharing with Firebase Cloud Storage, and a Laravel REST API backend managing posts, comments, and user profile data.
- Integrated Firebase Analytics for user behaviour tracking and ran structured in-app feedback sessions to iteratively improve the user experience based on what real users actually did.

DevelopersHub Applications | *DevelopersHub Corporation* | 2024 – 2025

- Architected and maintained multiple Flutter applications applying Clean Architecture principles throughout, ensuring each codebase remained navigable and extensible as features were added.
- Through deliberate and measured refactoring, reduced overall application sizes by 15% and improved load times by 25%, demonstrating the tangible impact of careful engineering decisions.

AI Nanny | *Google Hackathon* | 2024

As part of a team competing in the Google Hackathon 2024, I contributed to the design and development of AI Nanny, an AI-driven childcare assistant prototype.

- Defined user journeys, created wireframes and interactive prototypes in Figma, and built the frontend in Flutter with a particular focus on accessibility, including screen-reader support and scalable text rendering.
- Coordinated closely with machine learning specialists to integrate early-stage AI features including voice recognition and intelligent scheduling assistance.
- Received a Google Hackathon Participation Award for the team's work.

PORTFOLIO AND WORK LINKS

GitHub: <https://github.com/kazibwefrancis>

Hoscover: <https://hoscover-frontend-snowy.vercel.app>

SandiAI Web App: <https://sandi-ai-web-app-production.up.railway.app>

SandiAI Mobile App (Google Play): <https://play.google.com/store/apps/details?id=com.sandiai.app>

PACT Web Application: <https://pactorg.com>

PACT App Test Environment: <https://apptest.pactorg.com>

PACT Deployed App: <https://app.pactorg.com>

PACT Mobile App (Google Play): <https://play.google.com/store/apps/details?id=com.pact.workflow>

UCAA Survey System: <https://ucaa-survey.hippozug.com>

EDUCATION

Bachelor of Science in Software Engineering, Makerere University (In progress)

Uganda Advanced Certificate of Education, Viva College School (2020–2022)

Uganda Certificate of Education, Viva College School (2016–2019)

RECOGNITION AND ACHIEVEMENTS

- Innovation Consortium Medal of Excellence Award, Finalist in 2024 and 2025.
- Google Hackathon Participation Award, 2024, for the design and development of the AI Nanny prototype.
- PLP Africa Certificate in Web Development, 2023.

REFERENCES

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